MIT | Academy of Engineering

School of Computer Engineering
(Software Engineering)

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AIVY : An AI-Powered Gamified Learning Platform to Enhance Student Engagement and Personalised Education

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Introduction

AlVY: Revolutionizing Learning with AI & Gamification

The Problem

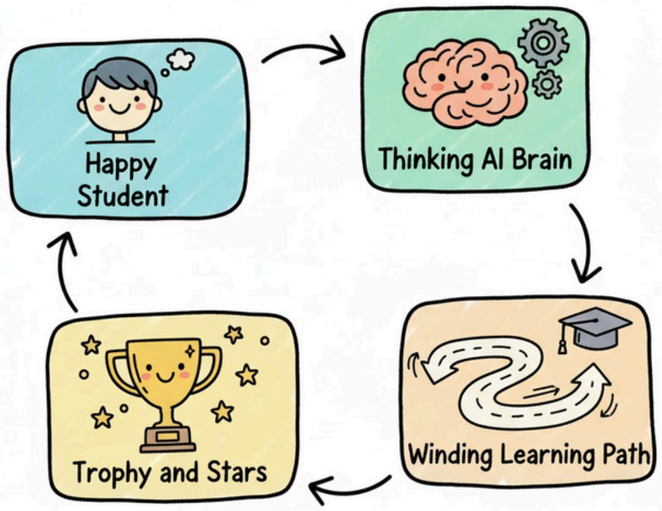
E-learning fails to engage students
Static, one-size-fits-all content
Poor completion rates due to lack of motivation

Our Solution

Dynamic platform merging AI + Gamification for personalized, engaging learning

How It Works

AI Personalization → Real-time performance analysis
Adaptive Paths → Automatic difficulty adjustment
Gamified Engagement → Points, badges, leaderboards
Continuous Feedback → Data refines AI model



Problem Statement

Traditional e-learning treats every student identically, delivering static content that fails to adapt to individual needs, which leads to widespread disengagement and ineffective learning outcomes.

Objectives and Scope

Objectives

- Develop an AI-powered, student-centered platform for personalized learning
- Integrate gamification & storytelling quizzes for interactive studying
- Enable AI-driven avatars & characters to guide learners
- Provide real-time feedback via video explanations & analytics
- Support self-awareness with session-wise SWOT analysis

Objectives and Scope

Scope

- Gamified Learning: Story-based quizzes, rewards & interactive avatars
- AI Personalization: Adaptive guidance & personalized study roadmaps
- Content Delivery: Dynamic video generation & character-based instructions
- Analytics & Reporting: Visual charts & SWOT feedback
- User Experience: Engaging, student-friendly UI with guided navigation

Need & Significance of This Study

Bridging the Gap in Modern Education

The Critical Problem: Why Current Systems Fail

- The Engagement Crisis: Over 85% of students report feeling disengaged in digital learning environments, leading to completion rates below 40% in most online courses.
- The Personalization Gap: Traditional e-learning platforms use a "one-size-fits-all" model, failing the 70% of learners who have unique paces and styles, leaving them behind or bored.
- The Motivation Drop: Without intrinsic motivation and real-time rewards, student effort and knowledge retention plummet over time.

Need & Significance of This Study

Bridging the Gap in Modern Education

AIVY's Unique Significance: The Integrated Solution

- **Synthesizes Two Proven Fields:** Unlike other platforms that use AI OR gamification in isolation, AIVY is novel in its deep integration of both to create a truly adaptive and motivating ecosystem.
- **Data-Driven Empathy:** Moves beyond static content to a system that learns from the student, creating a personal tutor-like experience that understands strengths and weaknesses.
- **Proactive Intervention:** Provides real-time SWOT analysis after every session, turning abstract performance data into a clear, actionable roadmap for improvement.
- **Builds a Growth Mindset:** The gamified elements are designed not just for short-term rewards, but to foster long-term resilience and a love for learning.

Need & Significance of This Study

Bridging the Gap in Modern Education

AIVY's Unique Significance: The Integrated Solution

For Learners:

- ↑ 60%+ Engagement through interactive storytelling and AI avatars.
- ↑ 45% Knowledge Retention via continuously adaptive learning paths.
- Empowerment through self-awareness from instant SWOT insights.

For Educational Institutions:

- A scalable tool to provide personalized attention to every student, regardless of class size.
- Actionable analytics for teachers to understand classroom trends and individual needs.
- For the Future of Ed-Tech:
- Establishes a new benchmark for ethically designed, student-centric AI that prioritizes fairness and genuine understanding over mere metrics.

Applications of Project management tools

Education
- Personalized student paths
- Teacher analytics & tracking

E-Learning
- Gamification & storytelling
- AI content generation

Exam Prep
- Gamified quizzes & AI avatars
- Targeted SWOT analysis

Self-Learning
- Adaptive goal-setting
- Skills & hobby roadmaps

Corporate
- AI-driven training sessions
- Progress reporting & upskilling

Special Ed
- Tailored pathways
- Accessible character guidance

Hardware/Software Requirement

Frontend Layer

- **Framework:** React.js / Next.js
- **Styling:** Tailwind CSS
- **Animations:** Framer Motion
- **Visualization:** Chart.js

AI & Machine Learning

- **NLP:** OpenAI / Hugging Face
- **Video Generation:** D-ID / Runway
- **Personalization:** TensorFlow
- **Speech:** Google Speech API

Backend & Database

- **Server:** Node.js / Django
- **Database:** PostgreSQL / MongoDB
- **Authentication:** Firebase Auth
- **Real-time:** Socket.io

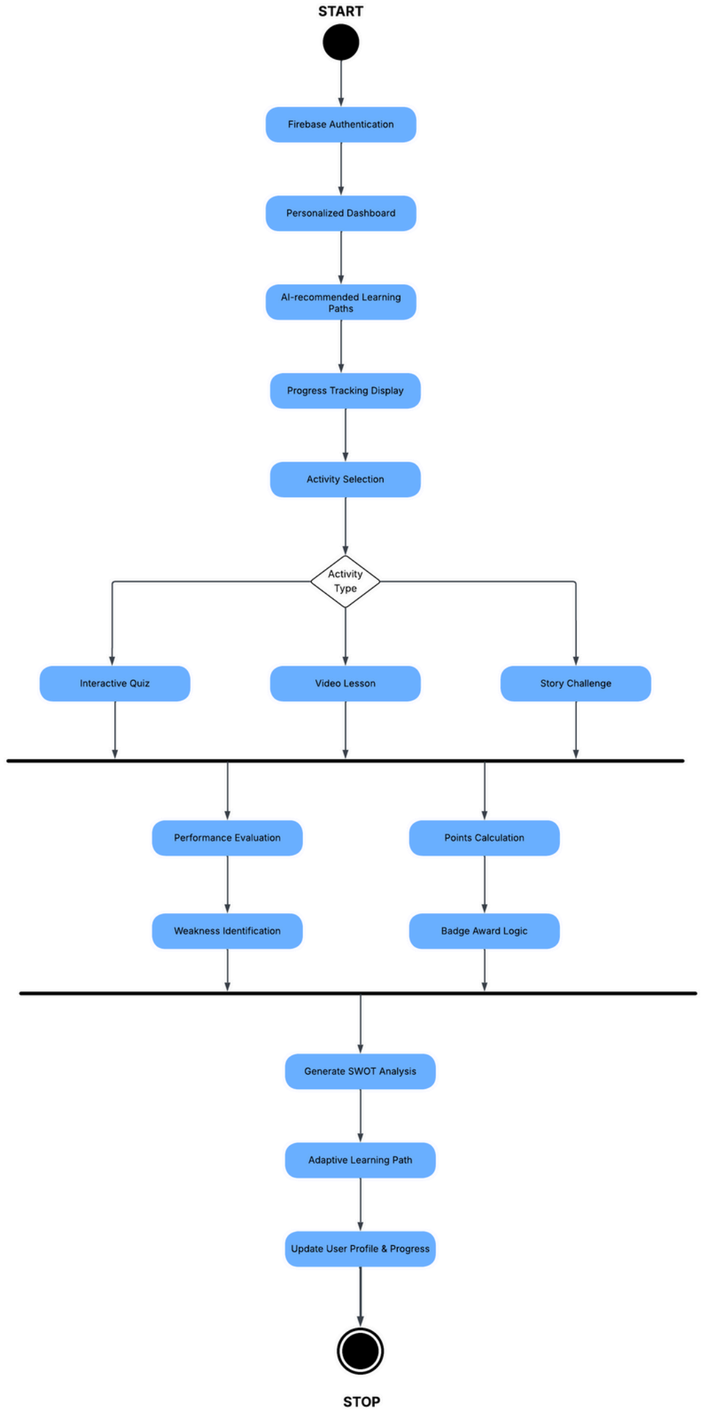
APIs & Services

- Custom Gamification API
- Google Charts Analytics
- Firebase Cloud Messaging

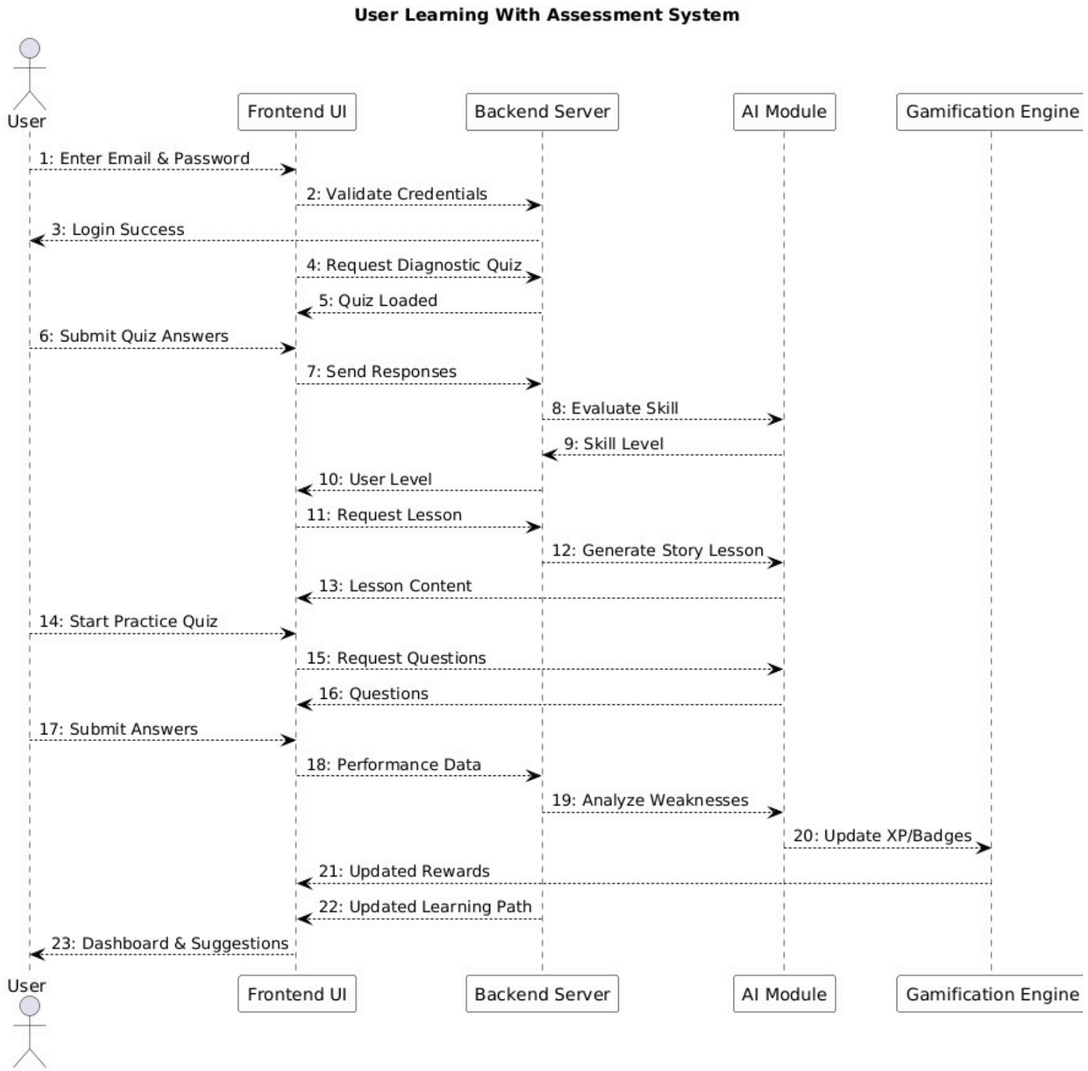
Literature Survey

Sr. No	Paper Title	Publication year and details	Authors	Technology Used	Features
1.	**The Impact of AI-Powered Gamified Learning Tools on Language Proficiency and Student Engagement of JHS Students from Private Schools in Bulacan**	2025, AJET, 4(3), 41–52	J.M.A. Dela Cruz, J.M. Santos J. Yambao	AI-powered gamified learning, digital platforms	Enhances language proficiency and engagement; includes teacher training and curriculum redesign
2.	**Artificial intelligence-enabled adaptive learning platforms**	2025, ScienceDirect	Le Ying Tan a, Shiyu Hu b, Darren J. Yeo c, Kang Hao Cheong	Adaptive AI algorithms, gamification	Provides adaptable learning experiences; supports multiple disciplines and ethical guidelines
3.	**Exploring the Impact of Gamification and AI on Personalized Educational Outcomes**	2025, IJRPR, Vol 6, Issue 4, 5543–5548	Gurunath Jagdhane, Tejal Bhosale	Gamification (points, badges), AI-driven personalization	Customizes learning paths; integrates interactive elements like badges and points; considers ethics & privacy
4.	**Does Gamification Work?—A Literature Review of Empirical Studies on Gamification**	2014, 47th HICSS	J. Hamari, J. Koivisto, H. Sarsa	Points, badges, leaderboards in ed tech	Reviews gamification effectiveness; analyses long-term engagement and ethical practices
5.	**Adaptive Gamification in E-learning: A Literature Review and Future Challenges**	2022, Comp. App. in Eng. Education, 30(2), 628–642	S. Bennani, A. Maalel, H. Ben Ghezala	Adaptive gamification, AI-based personalization	Balances motivation (intrinsic/extrinsic); supports sustainability and context transferability
6.	**Integrating AI and Gamification in Higher Education for E-Learning Optimization and Soft Skills Assessment**	2025, Frontiers in Computer Science, Vol 7, Article 1587040	Agostino Marengo, Alessandro Pagano, Brady Lund, Vito Santamato	AI (machine learning, LDA, LIME), gamified e-assessment, adaptive pathways	Provides adaptive assessment; improves data privacy; supports workforce and global accessibility

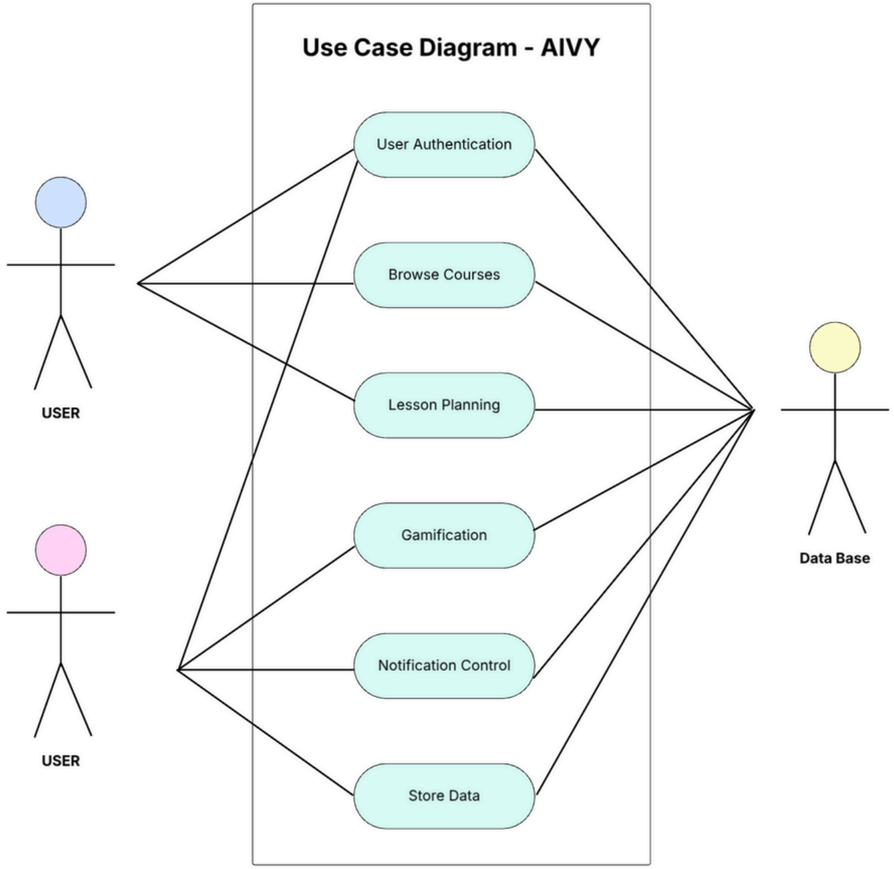
Algorithm/Activity Diagram



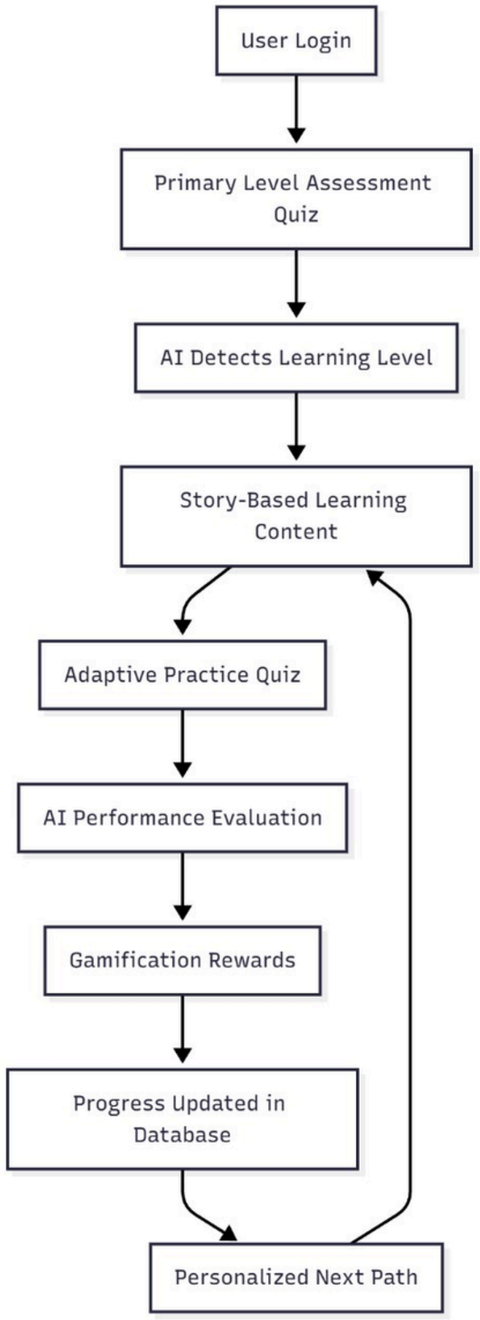
Sequence Diagram



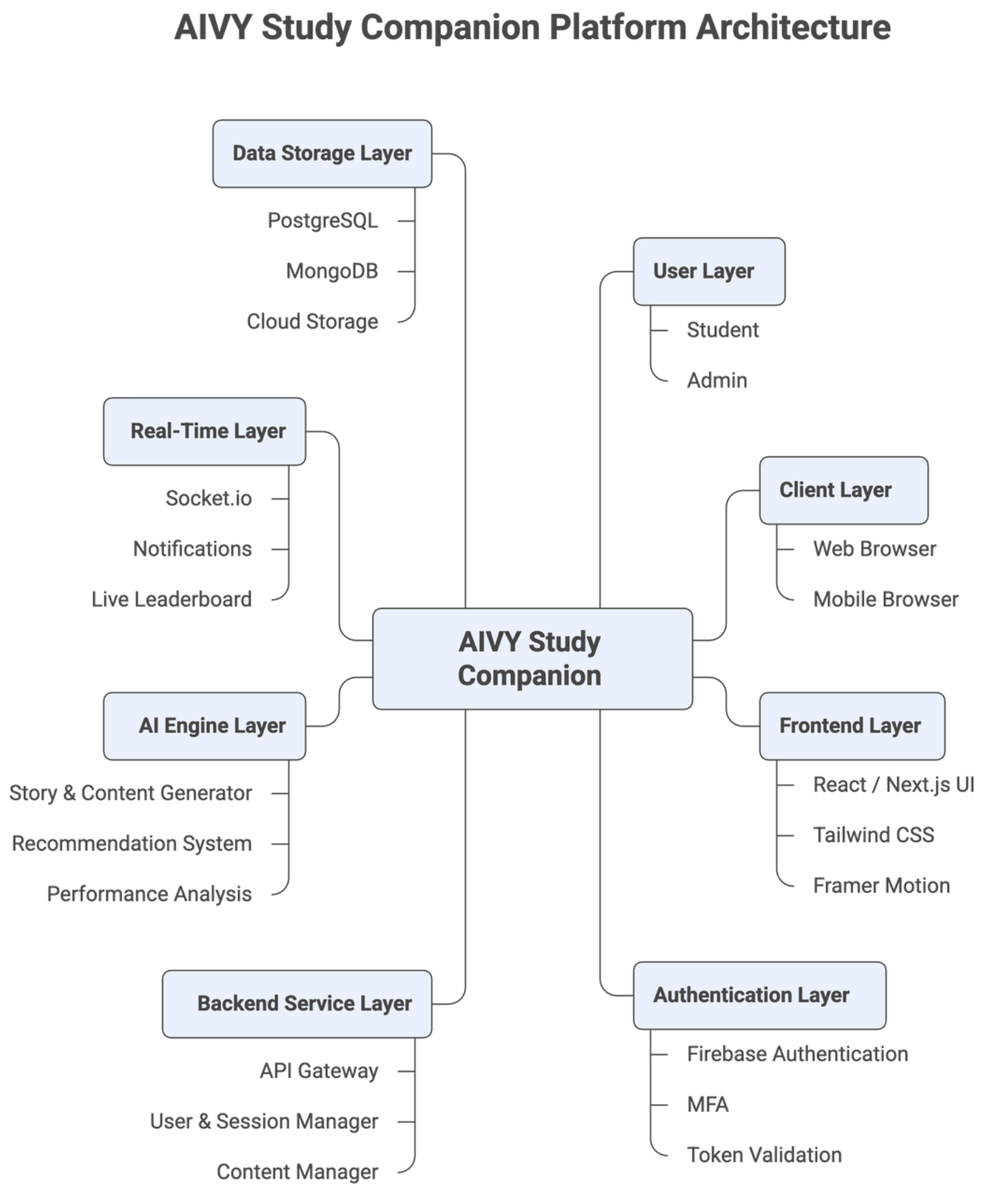
Use Case Diagram



System Architecture(Block)



System Architecture



Methodology

Methodology: Agile & Phased Development

1. Planning & Design Phase

- Requirement Analysis:
  - Tools: Jira, Confluence, Figma for user stories & documentation
  - Methods: User interviews, competitor analysis, feature prioritization matrix
- Architecture Planning:
  - Tools: Draw.io for system diagrams, AWS Architecture Center
  - Backend: Microservices architecture with Node.js/Express
- UI/UX Design:
  - Tools: Figma for prototyping, Adobe XD for wireframes
  - Framework: Material-UI component library with Tailwind CSS

2. Core Development Phase

- Secure Authentication:
  - Technology: Firebase Authentication with Multi-Factor Authentication (MFA)
  - Methods: SMS OTP, Google Authenticator integration, secure session management
- Basic Gamification Engine:
  - Backend: Custom Node.js API for points, badges, leaderboards
  - Database: PostgreSQL for user progress tracking
  - Real-time Updates: Socket.io for live leaderboard updates
- AI Quiz Engine:
  - Framework: TensorFlow.js for adaptive difficulty algorithms
  - APIs: Custom RESTful APIs for question management

Methodology

3. AI Integration Phase

- Personalized Learning Paths:
  - ML Framework: TensorFlow with recommendation algorithms
  - Data Processing: Python Pandas for learning analytics
  - APIs: Custom adaptive learning API with real-time adjustments
- AI Avatars & Video:
  - Avatar Creation: D-ID API for realistic AI presenter videos
  - Content Generation: OpenAI GPT-4 for script generation
  - Video Processing: FFmpeg for video optimization and streaming

4. Testing & Deployment Phase

- User Acceptance Testing:
  - Tools: Jest, Cypress for automated testing
  - Methods: A/B testing with different gamification elements
  - Platform: BrowserStack for cross-device compatibility testing
- Platform Deployment:
  - Frontend: Vercel for Next.js deployment
  - Backend: AWS EC2 instances with load balancers
  - Database: AWS RDS PostgreSQL with read replicas
  - Monitoring: New Relic for performance monitoring, Google Analytics for user behavior

Selected References

Academic Literature

[1] J. Hamari, J. Koivisto, and H. Sarsa, "Does Gamification Work? — A Literature Review of Empirical Studies on Gamification," in 2014 47th Hawaii International Conference on System Sciences, 2014, pp. 3025–3034.
DOI: 10.1109/HICSS.2014.377
(Foundational research on gamification effectiveness)

[2] A. Marengo, A. Pagano, B. Lund, and V. Santamato, "Integrating AI and Gamification in Higher Education for E-Learning Optimization and Soft Skills Assessment," Frontiers in Computer Science, vol. 7, Article 1587040, 2025.
DOI: 10.3389/fcomp.2025.1587040
(Recent peer-reviewed work connecting AI and gamification in education)

[3] G. Jagdhane and T. Bhosale, "Exploring the Impact of Gamification and AI on Personalized Educational Outcomes," Universal Review of Pedagogical Research, vol. 6, no. 4, pp. 5543–5548, 2025.
(Directly relevant to the project's core topic)

[4] S. Bennani, A. Maalel, and H. Ben Ghezala, "Adaptive Gamification in E-Learning: A Literature Review and Future Challenges," Computer Applications in Engineering Education, vol. 30, no. 2, pp. 628–642, 2022.
DOI: 10.1002/cae.22477
(Comprehensive review of adaptive gamification in e-learning).

Technical Documentation & Frameworks

[5] React.js Documentation. [Online]. Available: https://reactjs.org/docs/getting-started.html

[6] TensorFlow.js Documentation. [Online]. Available: https://www.tensorflow.org/js

[7] Firebase Authentication Documentation. [Online]. Available: https://firebase.google.com/docs/auth

[8] OpenAI API Documentation. [Online]. Available: https://platform.openai.com/docs/api-reference

Thank You!!
Any questions.